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Dear Editors,

On behalf of all co-authors (Rodrigo Almeida de Oliveira, Juliano de Paulo Ribeiro, and Prof. Edson Emilio Scalabrin), I am pleased to submit our manuscript titled **"Process Mining and Stochastic Modeling for Software Process Forecasting: A Systematic Mapping Study with a Human-Checked LLM-Assisted Screen"** for consideration in *Information and Software Technology*.

Novelty and contribution. This manuscript reports a systematic mapping study at the intersection of process mining (PM), stochastic modeling, and machine learning applied to software process forecasting. The contribution is threefold:

1. **Methodological:** a pre-registered, two-rater measurement of the selection error of an LLM-assisted screen (Claude Haiku 4.5 as primary screener, Claude Sonnet 4.6 as cross-model verifier). Against a pre-registered bar of Recall ≥ 0.90, Recall is 0.630 on the working-set sample (29 of 46) and 0.725 on the auxiliary tier after the stratum re-screen (235 of 324). The paper names the three mechanisms behind the shortfall (an abstract-based second pass, with a PDF read for 0.5% of the 886 records; records without an abstract closed by rule; and 32 of 338 coded includes, 9.5%, whose title or abstract does not support inclusion, which are removed) and reports the human re-screen the rule triggered. The cross-model κ (0.695 at title/abstract, 0.694 at the second pass) is reported as a secondary signal, and the round-1 FT row (Recall 0.246; $\kappa = 0.122$) is withdrawn.
2. **Empirical:** a corpus of 456 studies at the intersection of process mining, stochastic modelling, and software process forecasting: 306 that remain after the 32 removals from the list of 338, plus the 150 added studies (ADD-150) that remain after an eligibility pass removed 18 of 168. The studies are classified into four technique and data families and into integration levels L0–L3, with counts reported on named cuts (WS-151, the 151 working-set studies inside the 306, and the 293-study joint cut). One study in the 380 with at least one family mark (PRIMAD) matches process mining, a stochastic model, and forecasting. Two free-text fields (F2, F5) are withdrawn after a 0% exact match with a human rater on 31 studies; a single scope paragraph states what the counts describe.
3. **Synthetic:** the SPMF taxonomy and a four-direction research agenda (RA1–RA4), proposed by the authors. PATHCAST, the authors' own pipeline, is named as future work and is not evaluated.

Fit with IST scope. The manuscript squarely fits IST's scope on empirical software engineering methods, mining software repositories, and process modeling. It complements the journal's

growing body of evidence-based secondary studies while contributing to the methodological literature on LLM-assisted reviews.

Reproducibility. A complete replication package is openly available at Zenodo (DOI 10.5281/zenodo.21939600), including all search strings, screening prompts, JSON schemas, decision CSVs at every pipeline stage, the 456-study included list (`included_studies_506.csv`), the cross-model κ scripts, and the structured extraction tables.

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Prior submission and conflicts. The manuscript is original work, has not been published elsewhere, and is not under consideration by any other journal. All authors declare no competing financial interests. The use of generative AI tools in the research is declared in the manuscript per Elsevier policy. AI tools are *not* listed as authors and bear no responsibility for editorial judgements.

Suggested reviewers.

- **Wil M. P. van der Aalst** — RWTH Aachen University, foundational PM researcher (`wvdaalst@pads.rwth-aachen.de`)
- **Maria Teresa Baldassarre** — University of Bari, empirical SE and secondary studies
- **Eugene Syriani** — University of Montreal, LLM-assisted SLR methodology
- **Tore Dybå** — SINTEF / University of Oslo, evidence-based SE

Reviewers to avoid. None on basis of co-authorship within the past five years. Specific exclusions: thesis advisors and committee members at PUCPR.

Sincerely,

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